```
[0]%%sql TYPE MP
[0]%%sql DROP DATA
[0]%%sql DROP DATA
[0]%%sql DELETEFROM MP_PT
[0]%%sql DELETEFROM MP_PT_2
[0]%%sql DELETEFROM MP_PT_3
[0]%%sql DELETEFROM MP_PT_4
[0]%%sql DELETEFROM MP_PT_5
[0]%%sql DELETEFROM MP_PT_6 WHERE
```

Post Lab Photon Stats

Prepared by Jeremiah Conway

Question 1: Experimental Set-up

We have 2 models to work with here.

In experiment 1 we measured both the photon count event signal as well as the detector dead time. The count event signal follows the PTL standard meaning that it can be in a one voltage state (low) OR in a high voltage state (HI). Our experiment set up is as follows:

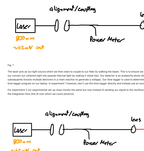


Fig 1

The dead zone up on light source which we then need to couple to a fiber by coupling the laser. This is to ensure we minimize the amount of light actually going into the fiber. We use the power meter to do this. The chopper is used to convert the continuous light source into a pulsed source. The pulse width is 20ns and the amplitude is 45mV. We then use the digital oscilloscope to view the output signal from the detector. The output signal from the detector is a TTL signal which is generated by a frosted glass which is placed in front of the detector. The frosted glass is used to diffuse the light so that it will be uniform across all our detector and ensure that we are not saturated. We also need to ensure that we are not saturated because we are using a photon counting detector. We want to be in the regime where we are counting individual photons.

For experiment 2 our experimental set up stays exactly the same but now instead of sending our signal to the oscilloscope we send it to the time tagger and then to a laptop. The relevant parameter for our LabVIEW interface is the experiment is the integration time (the # of count which we are collecting).

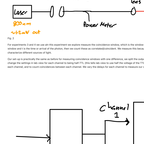


Fig 2

In experiment 2 and 3 we are all the experiment we explore measure the correlations among which is the ordered of time in which we consider 2 events to be correlated. In other words we count when is 1 detector event in the first detector and 1 in the second at most of the phase. We are also interested in the correlations between detectors. We measure the dead time of the detector using a PMT. We use the single output from the detectors to 2 input channels in the time tagger. We change the settings in the code for each channel to be the same as the settings of the PTL standard to match. Since we are counting our signal in both and also splitting the signal in half, the data collected will be half. We also need to count each channel, and to count correlations between each channel. We vary the delays for each channel to measure our correlations similar (we are looking for when our correlations counts drop to less ~10).

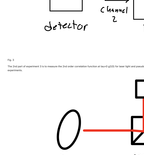


Fig 3

The first part of experiment 2 is to measure the 2nd order correlation function of light (G(2)) for laser light and pseudo thermal light. We do this by assembling a Hanbury Brown and Twiss interferometer, as our set up changes for the remaining experiments

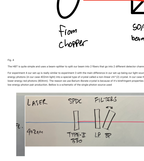


Fig 4

The HBT is quite simple and uses a beam splitter to split our beam into 2 fibers that go into 2 different detector channels so that we may count correlations.

Experiment 3 is the same as experiment 2 but we now introduce a rotating ground glass disk to our set up. The ground glass disk is used to simulate an approximate pseudothermal photon correlation. GPDC works by passing light through a rotating ground glass disk. Light passing through this material will be scattered such that it will exhibit a very linear cdf. Our goal is to use GPDC to a Bose-Einstein (BE) source. This source differentiates and allows for high photon bunching and spontaneously emit 2 beam bunge and photons differently. The reason we use Bartlett Bunch source is because of it's interferogram properties allowing us to do phase mastering which allows us to have the maximum conservation of visibility allowing us optimize the two energy photon correlation. Below is a schematic of the experimental set up:

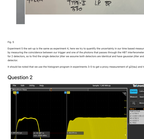
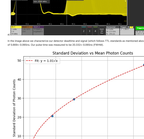


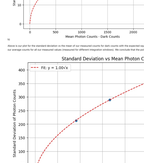
Fig 5

Experiment 5 the set up is the same as experiment 3, here we try to quantify the uncertainty in our time based measurements (aka known as detector jitter) which limits the temporal resolution we can achieve. We measure the jitter of the system using a pulsed laser source and a beam splitter. We use the time tagger to measure the time difference between the two channels. We then use the standard deviation of the distribution to determine the jitter of the system. For 5 experiments, we find that the single detector jitter we measured from experiment 5 is to get a survey measurement of groups and to characterize detector jitter.

Question 2



In the image above we characterize our detector deadtime and signal (which follows TTL standards as mentioned above). We measure a detector deadtime of 43.067±.0.000ns (our curve precision was ±0.0.000ns) and a rise time for our signal of 4.400ns ±.000ns. We also find the rise measurement to be 20.000±.0.000ns (FWHM).



fig

Below is a plot for the standard deviation vs the mean of our measured counts for various different intensities of our laser light with the expected squared model overlayed on top. What we see is that indeed the standard deviation matches very closely to the square root of our average counts for all our measurements for different intensities. We conclude that the uncertainty in our measurements follows a poisson distribution and is independent of intensity for photon count experiments.

Interesting to note here that we do not see an increase that is dependent on the intensity of our light source (i.e. we know that a poisonous source depends on photon count dependent).

As we mentioned in the intro, the measured histogram for experiment 3 combines the jitter of both detectors. To extract the timing jitter of a single detector, we can assume both models have the same detector jitter and we assume a gaussian model for both. We can use the histogram we have for both detectors in one quadrature by fitting it to a gaussian model. Then, we will to derive the combined gaussian to find a single gaussian. We do this by finding a relation between the parameters of our single gaussian model to that of our combined/convolution gaussian model. We can do this by using the convolution theorem and the characteristic function of the single gaussian. The convolution of 2 gaussians:

$$\mathcal{F}(f_{12}(x)) = \mathcal{F}(f_1(x)) \cdot \mathcal{F}(f_2(x)) = e^{-\frac{\sigma_1^2 t^2}{2}} \cdot e^{-\frac{\sigma_2^2 t^2}{2}}$$

https://mathworld.wolfram.com/Gaussian.html

Hence, our relation between our single detector standard deviations and combined detector standard deviation is simply:

$$\sigma_{12} = \sqrt{\sigma_1}$$

therefore the FWHM of a single detector's gaussian would be given by:

$$\sigma_{FWHM} = \sqrt{\frac{\sigma_{FWHM}^2}{2}} = 1.4 \cdot \frac{\sigma_{FWHM}}{\sqrt{2}}$$

Based on our model fit we get a single detector jitter of 0.980±.0.000ns

Question 3

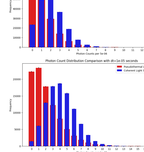
As the measured coincidence window of 43.067±.0.000ns, for coherent light we measured:

$$g(2) = 1.00 \pm 0.05$$

and from theory, we expect

$$g(2) = 1 + \frac{\sigma_{FWHM}^2 - \sigma_{det}^2}{\sigma_{FWHM}^2 - \sigma_{det}^2}$$

an our measurement match agree with the theory we expect.



fig

The Histogram shows that getting for coherent light is indeed centered on zero delay.

Question 4

a) For a chopper chaotic source of 1010±.0.000nm we measured:

$$g(2) = 1.50 \pm 0.05$$

For a purely thermal (see no source)

$$g(2) = 1 + \frac{\sigma_{FWHM}^2 - \sigma_{det}^2}{\sigma_{FWHM}^2 - \sigma_{det}^2}$$

Our light source of consist of an actual thermal light source, so we shouldn't expect a perfect value of 2. What we do expected however is photon bunching (g(2)) and we indeed do observe this.

fig

What the observe is a bunching of

$$g(2)$$

Because radiation spectra. This is because as we vary the time delay, the laser hits a different portion of the chopper, because the rotator chopper, as our calculations counts means for larger absorbate delays because the average between the portion of the chopper "rotor" to both channels is similar. The reason we have a gaussian shape is because we have a gaussian shaped beam, hence the center was hit both channel and also gaussian. For our calculated the histogram fit to a gaussian model, we find a FWHM of 1.000±.0.000ns. Using our relation from 3, we find a single detector jitter of 0.980±.0.000ns. We can use our single detector jitter to find our coherence time of our light source. We find that the coherence time is equal to $\tau_c = \frac{\sqrt{2}}{\sigma_{FWHM}} = 1.414 \cdot \frac{1}{\sigma_{FWHM}} = 1.414 \cdot \frac{1}{0.980 \pm 0.000}$ ns = 0.000±.0.000 ns. Our coherence time is a measure of a random source of photons. Because $\tau_c = \frac{1}{\Delta\nu} = 1.414 \cdot \frac{1}{\Delta\nu} = \frac{1}{0.000 \pm 0.000}$ = 0.000±.0.000 Hz. Unfortunately this means we cannot resolve the laser coherence spectra, we have higher standard deviation/coherence. I cannot assume that that the universal quantity is therefore related to the physical size of our laser beam/the area of projects to on the chopper. I'd have to do a bit more thinking to why and how this is the case but the coherence of our laser and the chopper/ground disk passes were dissipated the source to be less than coherent/photon bunch and get bunch and much closer at all.

Question 5

a) Below we show the time delayed values of all where our coherent light frequency histogram will looks consumer. Histograms are provided below for values of all = 1ns, 10ns, 20ns, 30ns and 100ns. We then plot the fit for fit our model to which this statistical fits a reasonable value.

b) Below we can see our count histogram generated for all bins for both coherent and pseudothermal light source. For coherent light we find $\alpha_{fitting}$ = 10.1±0.4 and $\alpha_{fitting}$ = 0.1±0.0001 for pseudothermal light we get $\alpha_{fitting}$ = 1.1±0.4 and $\alpha_{fitting}$ = 0.0002±0.0000 (I can not show the graphs with the uncertainties here. I need modified fitting and also to linearize the errors due to high the uncertainty in the chopper mark).

Question 6

a) $g(2) = 1.07 \pm 0.02$ for entangled light. For entangled light, we have a two mode squeezed state so $g(2)(\tau) = g(2)(\tau)$ which we do indeed have (while our value we measured would imply a thermal state this isn't the case here). I $\Delta\tau = 0.0002 \pm 0.0001$ ns and we find a FWHM of 0.000±.0.000ns.

b) For the triggered g(2) we measured g(2) = 0.0270 ± 0.0000 and we would expect a theoretical value of 0 for that looks like the theoretical $g(2) = 1 - \frac{1}{n}$. The reason we don't have a perfect value of 0 is because the 2 photon sources (or deep back states with n=1) leaked. The probability of getting states with n>1 is never zero so at times we will have back states with n>1 which will increase our measured g2 value. One way to reduce this source of error is to reduce the power of the laser to 1mW which ensures a source as this will effectively reduce the probability that we get states n>1 and. Another source of error could be inappropriate gain.

Appendix Code

```
%%sql DELETEFROM MP_PT_6;

%%sql SELECTFROM Import Data_PT
[0]%%sql SELECTFROM Import Data_Spectral
[0]%%sql SELECTFROM Import Data_Spectral_2 ORDER BY id DESC LIMIT 2000;

[0]%%sql SELECTFROM Import Data_Spectral_3 ORDER BY id DESC LIMIT 2000;

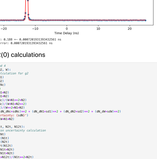
counts = pd.read_csv("counts.csv", delimiter = "\t", header = 0)
[0]%%sql SELECTFROM MP_PT_6 ORDER BY id DESC

pmt, pmt = scan_fit(spectrum, method='cauchy')[0'], ydata=counts['MP_6'], xlabel='Time', xlim=[-50, 5,2]

pmt.plot(legend=

%%sql
[0]%%sql SELECTFROM Import Data_PT
[0]%%sql SELECTFROM Import Data_Spectral_2
[0]%%sql SELECTFROM Import Data_Spectral_3
[0]%%sql SELECTFROM Import Data_Spectral_4, Import Data_Spectral_5 ORDER BY id DESC
pmt, pmt = scan_fit(spectrum, method='cauchy')[0'], ydata=counts['MP_5'], xlabel='Time', xlim=[-50, 5,2]
pmt.plot(legend=
pmt, pmt = scan_fit(spectrum, method='cauchy')[0'], ydata=counts['MP_4'], xlabel='Time', xlim=[-50, 5,2]
pmt.plot(legend=
pmt, pmt = scan_fit(spectrum, method='cauchy')[0'], ydata=counts['MP_3'], xlabel='Time', xlim=[-50, 5,2]
pmt.plot(legend=
pmt, pmt = scan_fit(spectrum, method='cauchy')[0'], ydata=counts['MP_2'], xlabel='Time', xlim=[-50, 5,2]
pmt.plot(legend=
pmt, pmt = scan_fit(spectrum, method='cauchy')[0'], ydata=counts['MP_1'], xlabel='Time', xlim=[-50, 5,2]
pmt.plot(legend=

[0]%%sql SELECTFROM Import Data_Spectral_6
[0]%%sql SELECTFROM Import Data_Spectral_7
[0]%%sql SELECTFROM Import Data_Spectral_8
[0]%%sql SELECTFROM Import Data_Spectral_9
[0]%%sql SELECTFROM Import Data_Spectral_10
[0]%%sql SELECTFROM Import Data_Spectral_11
[0]%%sql SELECTFROM Import Data_Spectral_12
[0]%%sql SELECTFROM Import Data_Spectral_13
[0]%%sql SELECTFROM Import Data_Spectral_14
[0]%%sql SELECTFROM Import Data_Spectral_15
[0]%%sql SELECTFROM Import
```



Scale Factor Inferred: 0.500 ± 0.000[0.0000]0000[0000] ns
Standard Deviation Error: 0.000[0]0000[0000] ns

Code for g(2)calculations

```
# Main execution
def main(file, N1, N2, N3):
    # Basic parameters calculation for g2
    N = N1 + N2 + N3
    n0 = N / (N+3)
    n1 = n0
    n2 = n0
    n3 = n0
    # Calculate g2 values
    g_11 = (N+1)/(N+3)
    g_22 = (N+1)/(N+3)
    g_33 = (N+1)/(N+3)
    g_12 = (N+1)/(N+3)+1/(N+3)
    g_13 = (N+1)/(N+3)+1/(N+3)
    g_23 = (N+1)/(N+3)+1/(N+3)
    print("g2 uncorrelated: {:.3f}".format(g_11))
    print("g2 correlated: {:.3f}".format(g_12))
    return (N,n0,n1,n2,n3,g_11,g_22,g_33,g_12,g_13,g_23)

def plot(g,N1,N2,N3,M1,M2):
    # Plot the g2 and fitting calculation
    ax = plt.subplots()[1]
    ax.set_ylim(0,2.5)
    ax.set_xlim(-0.5,0.5)
    ax.set_xlabel("Time Delay")
    ax.set_ylabel("g2")
    ax.set_title("g2 Fitting")
    ax.plot(g[0],'r',label="g2 Data")
    ax.plot(g[5],'b',label="g2 Fit")
    ax.plot(g[6],'g',label="g2 Fit")
    ax.plot(g[7],'y',label="g2 Fit")
    ax.legend(loc="upper right")
    ax.text(0.05,0.95,"N1: {:.0f}\nN2: {:.0f}\nN3: {:.0f}\nM1: {:.0f}\nM2: {:.0f}".format(N1,N2,N3,M1,M2),transform=ax.transform_axes,fontsize=10,verticalalignment="top")
    plt.show()

# Main execution
if __name__ == "__main__":
    main(1000,1000,1000,1000,1000)
```

Code for g2(tau) histograms

```
# Histogram plot for coherent light and pseudothermal light bunches starts on row 20
coherent = np.read_csv("C_coherent.txt", delimiter=" ",header=None)
pseudothermal = np.read_csv("C_pseudothermal.txt", delimiter=" ",header=None)
pseudothermalLOW = np.read_csv("C_pseudothermalLOW.txt", delimiter=" ",header=None)
pseudothermalHIGH = np.read_csv("C_pseudothermalHIGH.txt", delimiter=" ",header=None)
# Accounts first 10 data points in time series
coherent = coherent[coherent[0]>=10000]
pseudothermal = pseudothermal[pseudothermal[0]>=10000]
pseudothermalLOW = pseudothermalLOW[pseudothermalLOW[0]>=10000]
pseudothermalHIGH = pseudothermalHIGH[pseudothermalHIGH[0]>=10000]

# Fit linear model to coherent data

def linear(x, a, b):
    return a * x + b

fit_linear_params, covariance = curve_fit(linear, coherent[0].to_numpy(), coherent[1].to_numpy(), p0=[0, np.mean(coherent[1].to_numpy())])


# Fit linear model to coherent data

plt.figure(figsize=(16,8))
plt.scatter(coherent[0],coherent[1],s=1,alpha=0.5,
plt.plot(coherent[0],linear(coherent[0],*fit_linear_params),color="red",lw=2,label="Linear Fit")
plt.title("Histogram of Two-Photon Coincidence Counts for Coherent Light Source")
plt.show()


# Fit gaussian to pseudothermal data

def gaussian(x, amplitude, mean, sigma, baseline):
    return amplitude * np.exp(-(x - mean) ** 2) / (2 * sigma ** 2) + baseline

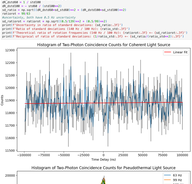
sigmaLOW = np.std(np.abs(pseudothermalLOW[0]))
meanLOW = np.mean(np.abs(pseudothermalLOW[0]))
sigmaLOW = np.std(np.abs(pseudothermalLOW[0]))

LOW_guess_mean = np.mean(np.abs(pseudothermalLOW[0]))
LOW_guess_amplitude = np.max(pseudothermalLOW[1])-np.min(pseudothermalLOW[1])
LOW_guess_sigma = np.std(np.abs(pseudothermalLOW[0]))
LOW_guess_baseline = np.min(pseudothermalLOW[1])
fit_gaussian_LOW_params, covariance = curve_fit(gaussian, pseudothermalLOW[0].to_numpy(), pseudothermalLOW[1].to_numpy(), p0=[LOW_guess_amplitude, LOW_guess_mean, LOW_guess_sigma, LOW_guess_baseline])

# Plot gaussian fit to pseudothermal data
plt.figure(figsize=(16,8))
plt.scatter(pseudothermalLOW[0],pseudothermalLOW[1],s=1,alpha=0.5,
plt.plot(pseudothermalLOW[0],gaussian(pseudothermalLOW[0],*fit_gaussian_LOW_params),color="orange",lw=2,label="LOW Fit")
plt.title("Histogram of Two-Photon Coincidence Counts for Pseudothermal Light Source")
plt.show()

# Plot histogram for all of coherent deviation between the 100 and 100 bit fits
# Coherent deviation
diff = fit_linear_params[1]
ratio_list = diff/fit_linear_params[1]

# Normalizing calculation for ratio
ad_diff = np.ptp(fit_linear_params[1])
ad_ratio = ad_diff/fit_linear_params[1]
ad_diff = ad_diff
ad_ratio = ad_ratio
ad_ratio = np.ptp(LOW_diff/LOW_ratio_list)
LOW_diff = LOW_diff
LOW_ratio = LOW_ratio
ad_ratio = np.ptp(LOW_diff/LOW_ratio_list)
ad_ratio = ad_ratio
print("Peak-to-Peak Amplitude Ratio: {:.3f}".format(ad_ratio))
print("Standard Deviation of Ratio: {:.3f}".format(np.std(ratio_list)))
print("Percentage of ratio of standard deviations: {:.3f}".format(np.std(ratio_list)/ad_ratio))
```




```
Uncertainty in ratio of standard deviations: 0.015
Peak-to-Peak Amplitude Ratio: 0.000[0]0000[0000]
Theoretical ratio of residual fluctuations (100 Hz / 100 Hz): 1.575 ± 0.014
Percentage of ratio of standard deviations: 0.010 ± 0.001
```

Code for convolution fitting

```
# Load/Save data
coherent = pd.read_csv("coreTWO.txt", delimiter=" ",header=None, names=["Channel", "Time"])
pseudothermal = pd.read_csv("pseudothermalTWO.txt", delimiter=" ",header=None, names=["Channel", "Time"])

# Calculate time difference between channels

def get_c(series, bins):
    return stats.poisson.pmf(r, mean)

def CheckN(r, mean):
    return 1/mean * 1/(r + 1)*mean ** r
    # Calculate the Poisson PMF

def pd_c(r, mean_p, mean_c):
    # Analogouse of poisson and thermal distributions at point r
    # r - bin
    # mean_p - mean of poisson
    # mean_c - mean of thermal (r - 1 because r is bin)
    return mean

def convolve_distributions(r, mean_p, mean_c):
    p = CheckN(r, mean_p) * pd_c(r, mean_p, mean_c)
    return p

def FitC(r, channel, df, mean_p,c=0, mean_c,c=0, lightsource):
    # Calculate the convolution of poisson and thermal, and fit it to data
    mean_p,c = df[df[channel].isin([r])].count()
    mean_c,c = df[df[channel].isin([r-1])].count()
    # Calculate the convolution
    local_freq = convolve_distributions(r, mean_p,c, mean_c,c)
    local_freq = r, local_freq
    return local_freq, lightsource

# Fitting with g2 and channel count, and normalize frequency to sum to 1
local_freq1.frequency = local_freq1.frequency / local_freq1.frequency.sum()
local_freq2.frequency = local_freq2.frequency / local_freq2.frequency.sum()

# Combine
local_freq1["frequency"] = local_freq1["frequency"].fillna(0)
local_freq2["frequency"] = local_freq2["frequency"].fillna(0)
local_freq = pd.concat([local_freq1, local_freq2])
local_freq["frequency"] = local_freq["frequency"] / local_freq["frequency"].sum()""
pdata = local_freq[local_freq[0].isin(r_range)]
pdata = pdata[pdata["lightsource"] == "coherent"]["frequency"].to_numpy(dtype=float)
chi2 = chisq(r_range, pdata, mean_p, mean_c, lambda r, mean_p,c, mean_c,c: convolve_distributions(r, mean_p,c, mean_c,c))
print("Chi2: {:.3f}".format(chi2))
plt.figure(figsize=(20,8))
sns.barplot(data=local_freq, x="bins", y="frequency", hue="channel", dodge=True)
plt.title("Photon Count Distribution for pseudothermal light source with df=2.5e-05 seconds")
plt.xlabel("Photon Count per 2.5e-05")
plt.ylabel("Normalized Frequency")
plt.legend(["Pd: Mean Poisson=1.22, Mean Thermal=0.78"])
plt.show()
print("Coherent Photon Counts per 1ns")
print(coherent[1].to_numpy())
plt.figure(figsize=(20,8))
plt.plot(coherent[1].to_numpy(),"o")
plt.title("Coherent Photon Count Distribution for 1ns")
plt.xlabel("Photon Count")
plt.ylabel("Frequency")
plt.show()

# Bright mode and their separations
bright = coherent[coherent[1].isin([1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27,28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43,44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59,60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75,76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91,92,93,94,95,96,97,98,99,100,101,102,103,104,105,106,107,108,109,110,111,112,113,114,115,116,117,118,119,120,121,122,123,124,125,126,127,128,129,130,131,132,133,134,135,136,137,138,139,140,141,142,143,144,145,146,147,148,149,150,151,152,153,154,155,156,157,158,159,160,161,162,163,164,165,166,167,168,169,170,171,172,173,174,175,176,177,178,179,180,181,182,183,184,185,186,187,188,189,190,191,192,193,194,195,196,197,198,199,200,201,202,203,204,205,206,207,208,209,210,211,212,213,214,215,216,217,218,219,220,221,222,223,224,225,226,227,228,229,230,231,232,233,234,235,236,237,238,239,240,241,242,243,244,245,246,247,248,249,250,251,252,253,254,255,256,257,258,259,260,261,262,263,264,265,266,267,268,269,270,271,272,273,274,275,276,277,278,279,280,281,282,283,284,285,286,287,288,289,290,291,292,293,294,295,296,297,298,299,300,301,302,303,304,305,306,307,308,309,310,311,312,313,314,315,316,317,318,319,320,321,322,323,324,325,326,327,328,329,330,331,332,333,334,335,336,337,338,339,340,341,342,343,344,345,346,347,348,349,350,351,352,353,354,355,356,357,358,359,360,361,362,363,364,365,366,367,368,369,370,371,372,373,374,375,376,377,378,379,380,381,382,383,384,385,386,387,388,389,390,391,392,393,394,395,396,397,398,399,400,401,402,403,404,405,406,407,408,409,410,411,412,413,414,415,416,417,418,419,420,421,422,423,424,425,426,427,428,429,430,431,432,433,434,435,436,437,438,439,440,441,442,443,444,445,446,447,448,449,450,451,452,453,454,455,456,457,458,459,460,461,462,463,464,465,466,467,468,469,470,471,472,473,474,475,476,477,478,479,480,481,482,483,484,485,486,487,488,489,490,491,492,493,494,495,496,497,498,499,500,501,502,503,504,505,506,507,508,509,510,511,512,513,514,515,516,517,518,519,520,521,522,523,524,525,526,527,528,529,530,531,532,533,534,535,536,537,538,539,540,541,542,543,544,545,546,547,548,549,550,551,552,553,554,555,556,557,558,559,560,561,562,563,564,565,566,567,568,569,570,571,572,573,574,575,576,577,578,579,580,581,582,583,584,585,586,587,588,589,590,591,592,593,594,595,596,597,598,599,600,601,602,603,604,605,606,607,608,609,610,611,612,613,614,615,616,617,618,619,620,621,622,623,624,625,626,627,628,629,630,631,632,633,634,635,636,637,638,639,640,641,642,643,644,645,646,647,648,649,650,651,652,653,654,655,656,657,658,659,660,661,662,663,664,665,666,667,668,669,670,671,672,673,674,675,676,677,678,679,680,681,682,683,684,685,686,687,688,689,690,691,692,693,694,695,696,697,698,699,700,701,702,703,704,705,706,707,708,709,710,711,712,713,714,715,716,717,718,719,720,721,722,723,724,725,726,727,728,729,730,731,732,733,734,735,736,737,738,739,740,741,742,743,744,745,746,747,748,749,750,751,752,753,754,755,756,757,758,759,760,761,762,763,764,765,766,767,768,769,770,771,772,773,774,775,776,777,778,779,780,781,782,783,784,785,786,787,788,789,790,791,792,793,794,795,796,797,798,799,800,801,802,803,804,805,806,807,808,809,810,811,812,813,814,815,816,817,818,819,820,821,822,823,824,825,826,827,828,829,830,831,832,833,834,835,836,837,838,839,840,841,842,843,844,845,846,847,848,849,850,851,852,853,854,855,856,857,858,859,860,861,862,863,864,865,866,867,868,869,870,871,872,873,874,875,876,877,878,879,880,881,882,883,884,885,886,887,888,889,890,891,892,893,894,895,896,897,898,899,900,901,902,903,904,905,906,907,908,909,910,911,912,913,914,915,916,917,918,919,920,921,922,923,924,925,926,927,928,929,930,931,932,933,934,935,936,937,938,939,940,941,942,943,944,945,946,947,948,949,950,951,952,953,954,955,956,957,958,959,960,961,962,963,964,965,966,967,968,969,970,971,972,973,974,975,976,977,978,979,980,981,982,983,984,985,986,987,988,989,990,991,992,993,994,995,996,997,998,999,1000,1001,1002,1003,1004,1005,1006,1007,1008,1009,1010,1011,1012,1013,1014,1015,1016,1017,1018,1019,1020,1021,1022,1023,1024,1025,1026,1027,1028,1029,1030,1031,1032,1033,1034,1035,1036,1037,1038,1039,1040,1041,1042,1043,1044,1045,1046,1047,1048,1049,1050,1051,1052,1053,1054,1055,1056,1057,1058,1059,1060,1061,1062,1063,1064,1065,1066,1067,1068,1069,1070,1071,1072,1073,1074,1075,1076,1077,1078,1079,1080,1081,1082,1083,1084,1085,1086,1087,1088,1089,1090,1091,1092,1093,1094,1095,1096,1097,1098,1099,1100,1101,1102,1103,1104,1105,1106,1107,1108,1109,1110,1111,1112,1113,1114,1115,1116,1117,1118,1119,1120,1121,1122,1123,1124,1125,1126,1127,1128,1129,1130,1131,1132,1133,1134,1135,1136,1137,1138,1139,1140,1141,1142,1143,1144,1145,1146,1147,1148,1149,1150,1151,1152,1153,1154,1155,1156,1157,1158,1159,1160,1161,1162,1163,1164,1165,1166,1167,1168,1169,1170,1171,1172,1173,1174,1175,1176,1177,1178,1179,1180,1181,1182,1183,1184,1185,1186,1187,1188,1189,1190,1191,1192,1193,1194,1195,1196,1197,1198,1199,1200,1201,1202,1203,1204,1205,1206,1207,1208,1209,1210,1211,1212,1213,1214,1215,1216,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